



Attribute and Simile Classifiers For Verification

Neeraj Kumar Alexander C.Berg Peter N.Belhumeur
Shree K. Nayar . Columbia University

Motivation

- The difficulty for face verification:
- Enormous variability of a face for a camera:
- Poses, Expression, Hairstyle
- Illumination direction, camera type, focus, resolution, image compression

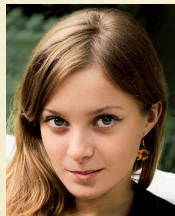
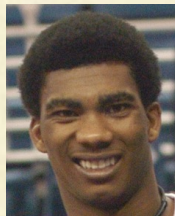
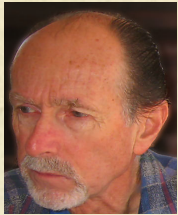
Attributes can define categories

Female

Eyeglasses

Middle-aged

Dark hair



Kumar, C.Berg, N.Belhumeur and Nayar. "Attribute and Similarity Classifiers For Verification"

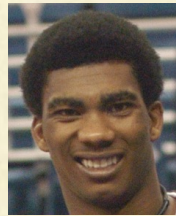
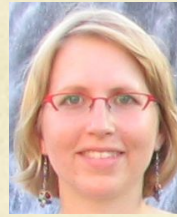
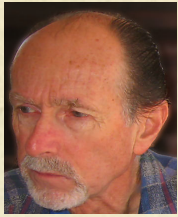
Attributes can define categories

Caucasian

Teeth showing

Outside

Tilted head



Kumar, C.Berg, N.Belhumeur and Nayar. "Attribute and Simile Classifiers For Verification"

Are these images of the same person?



Kumar, C.Berg, N.Belhumeur and Nayar. "Attribute and
Simile Classifiers For Verification"

Prior approaches

Images → Low-level features → Verification



→
RGB
HOG
LBP
SIFT
...



→
RGB
HOG
LBP
SIFT
...



Different

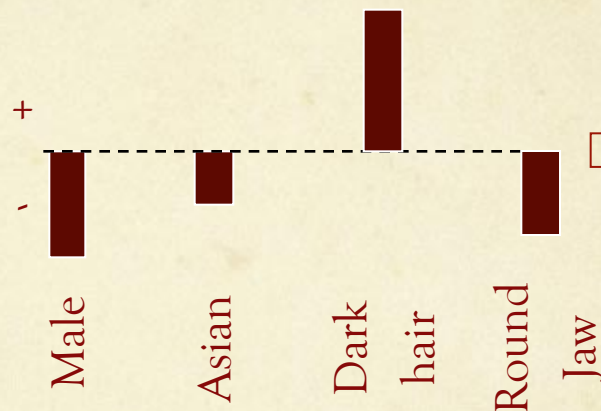
Our approach: attributes



RGB
HOG
LBP
SIFT
...



RGB
HOG
LBP
SIFT
...



Different

Core Idea

- Extracting and comparing “high-level” visual features or traits of a face image
- Insensitive to image conditions
- Vector of outputs from the trait classifiers are computed on each face independently

“Attribute” classifier

- Use binary classifier trained to recognize the presence or absence of describable aspects of visual appearance
- Gender, age, nose type, skin, race, hair color, eye type

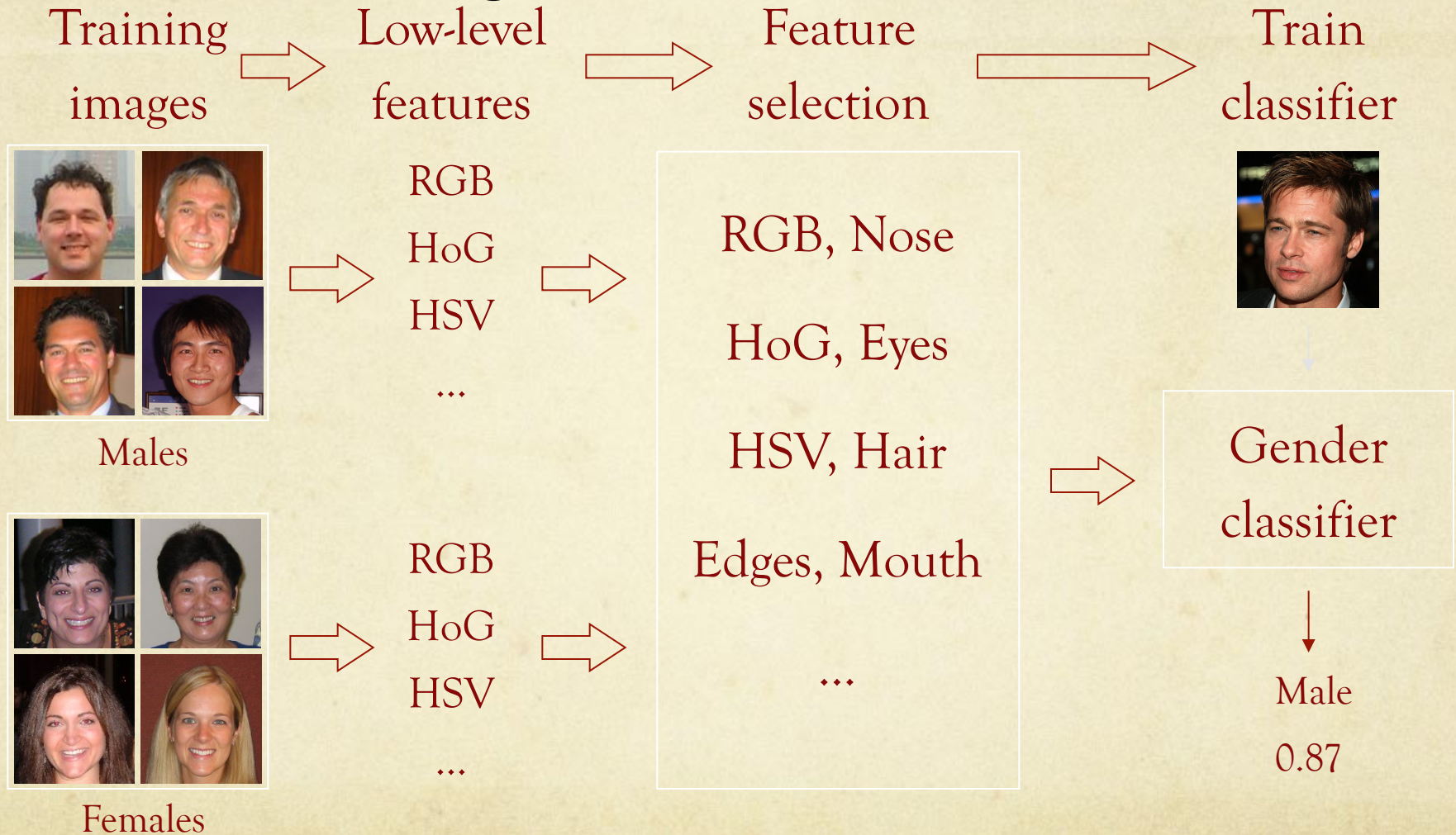
“Simile” classifier

- Removes the manual labeling required for attribute classification
- Instead, simile classifier learns the similarity of faces, or regions of faces, to specific reference people.

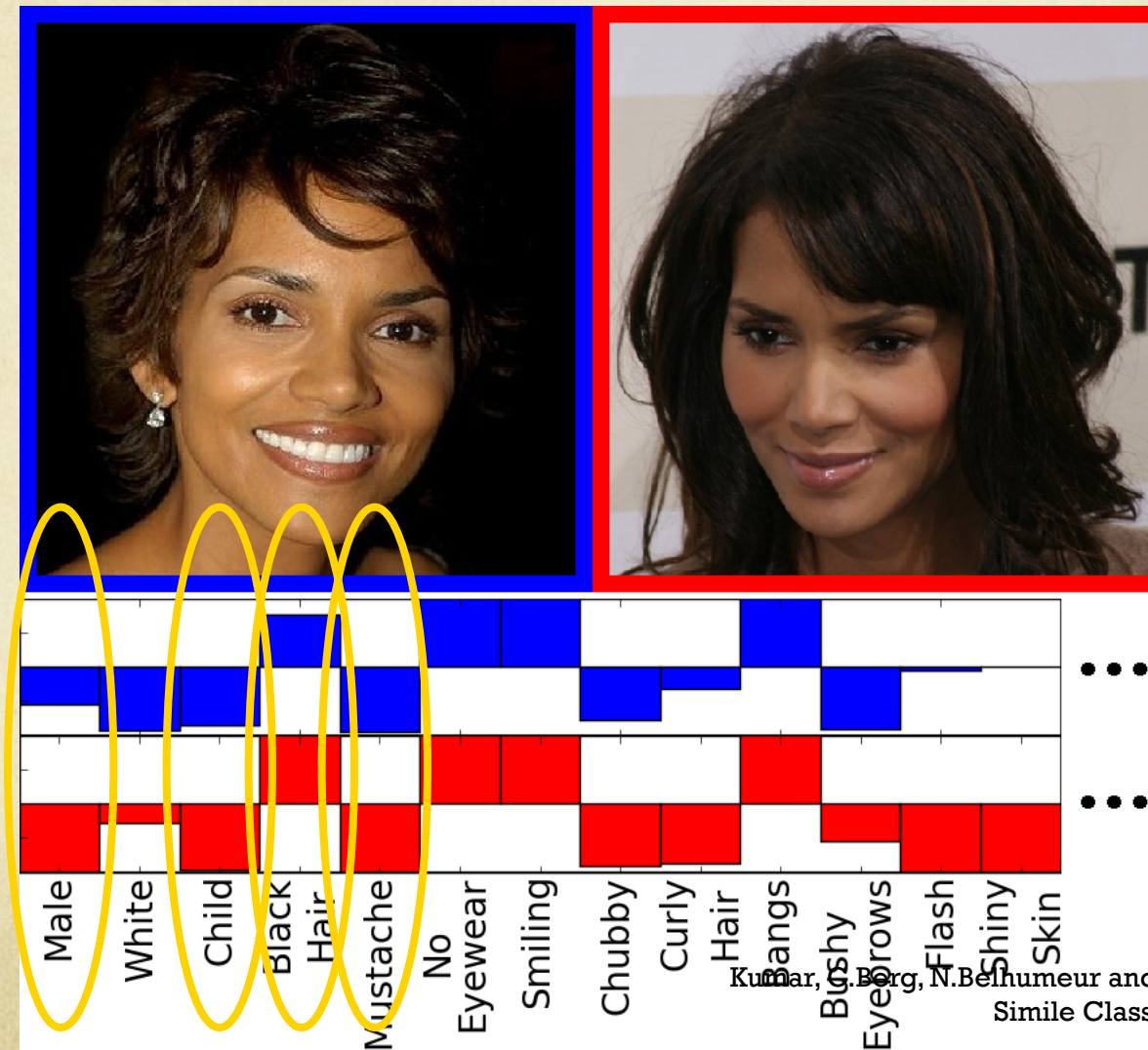
Approach

- Extract Low-Level Features
- Compute Visual Traits
- Perform Verification

Learning an attribute classifier



Use attributes to perform verification



Verification
classifier

Kuhlbar, J. Borg, N. Belhumeur and Nayar. "Attribute and Simile Classifiers For Verification"

Attributes are intuitive

Female

Young

Attractive

White



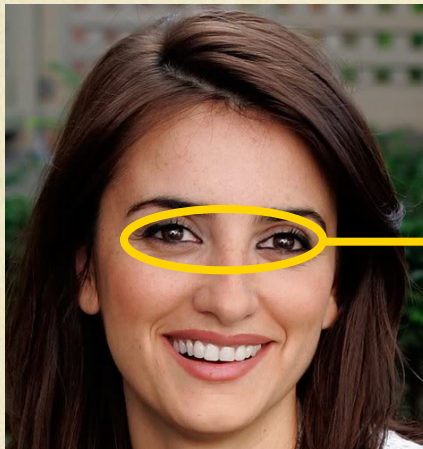
Black hair

Frontal pose

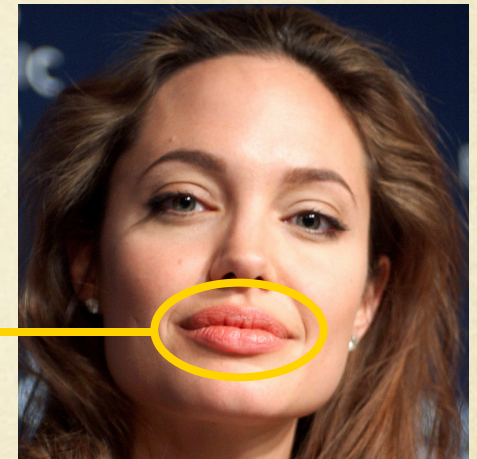
Mouth closed

Eyes open

Describe faces using similes

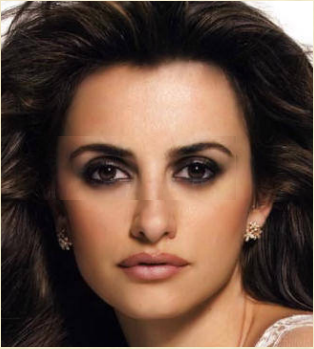


Penelope Cruz

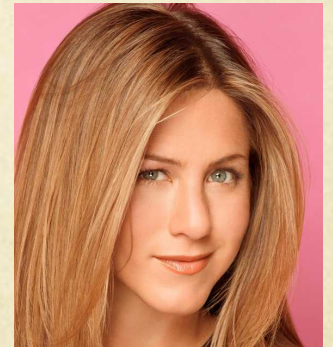
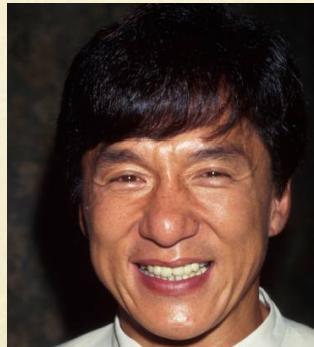


Angelina Jolie

Training simile classifiers



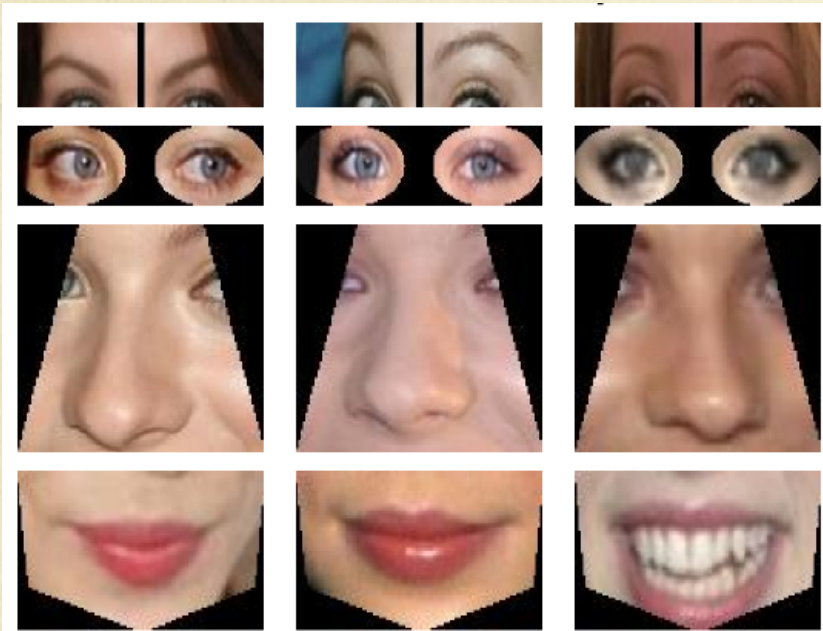
Images of Penelope Cruz's eyes



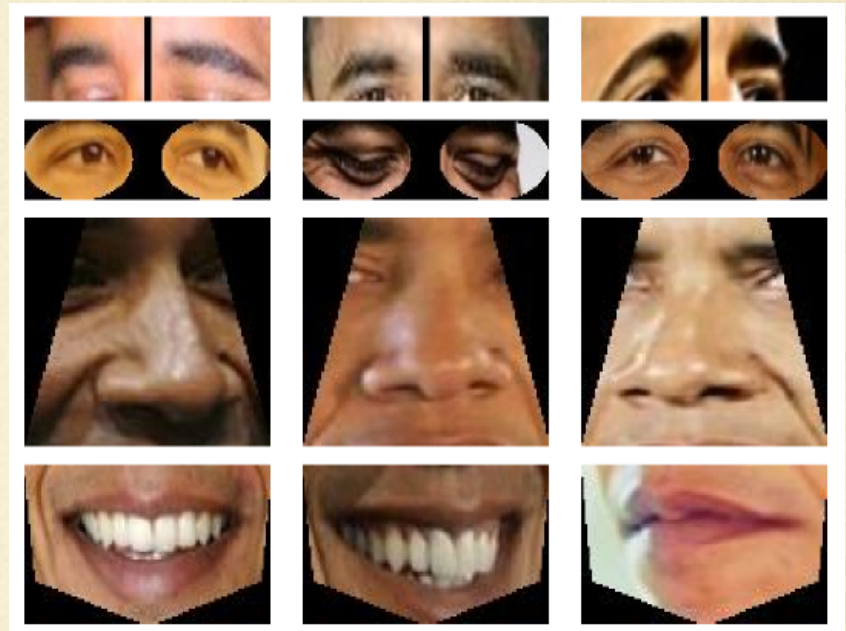
Images of other people's eyes

Similes for training

Positive Examples

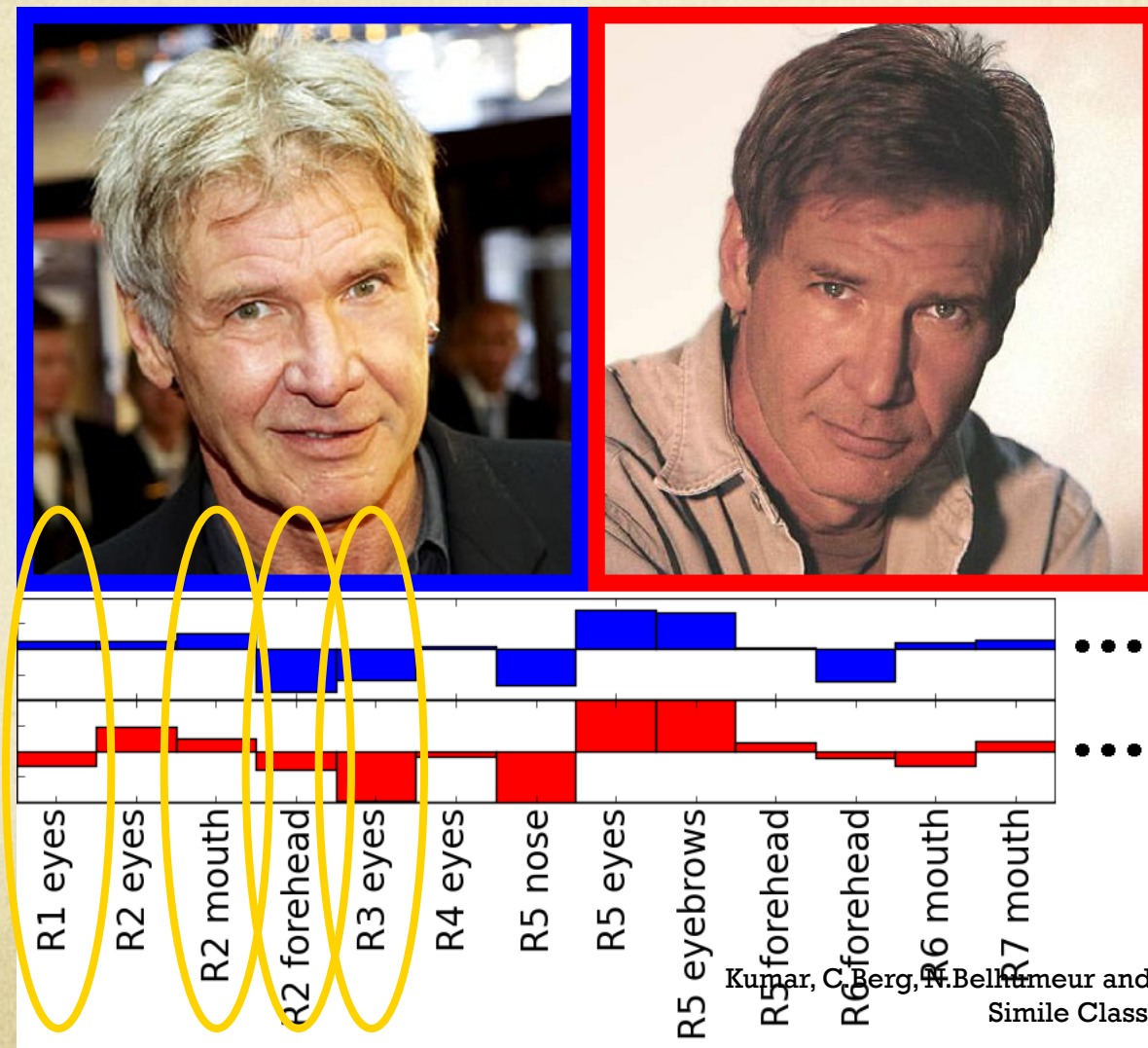


Reference Person R1



Reference Person R2

Using simile classifiers for verification



Verification
classifier

Experiment

- Evaluate performance on a face verification task: given two images of faces, determine if they show the same individual
- No overlaps in the individuals used in training and testing
- LFW and PubFig Benchmark

Labeled Faces in the Wild (LFW)

Labeled Faces in the Wild



Menu

- LFW Home
- UMass Vision

Database by name, non-singleton

[A][B][C][D][E][F][G][H][I][J][K][L][M][N][O][P][Q][R][S][T][U][V][W][X][Y][Z]



Habib Rizieq (5)



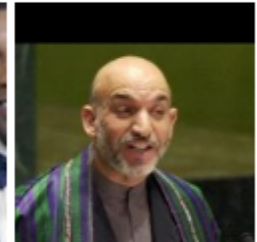
Hal Gehman (5)



Hal Sutton (2)



Halle Berry (16)



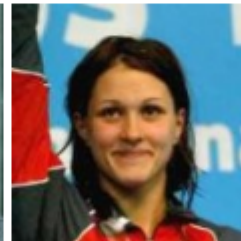
Hamid Karzai (22)



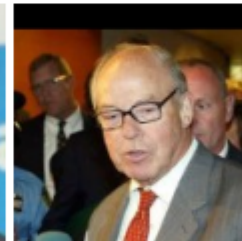
Hamzah Haz (2)



Hanan Ashrawi
(2)



Hannah
Stockbauer (2)



Hans Blix (39)



Hans Eichel (3)

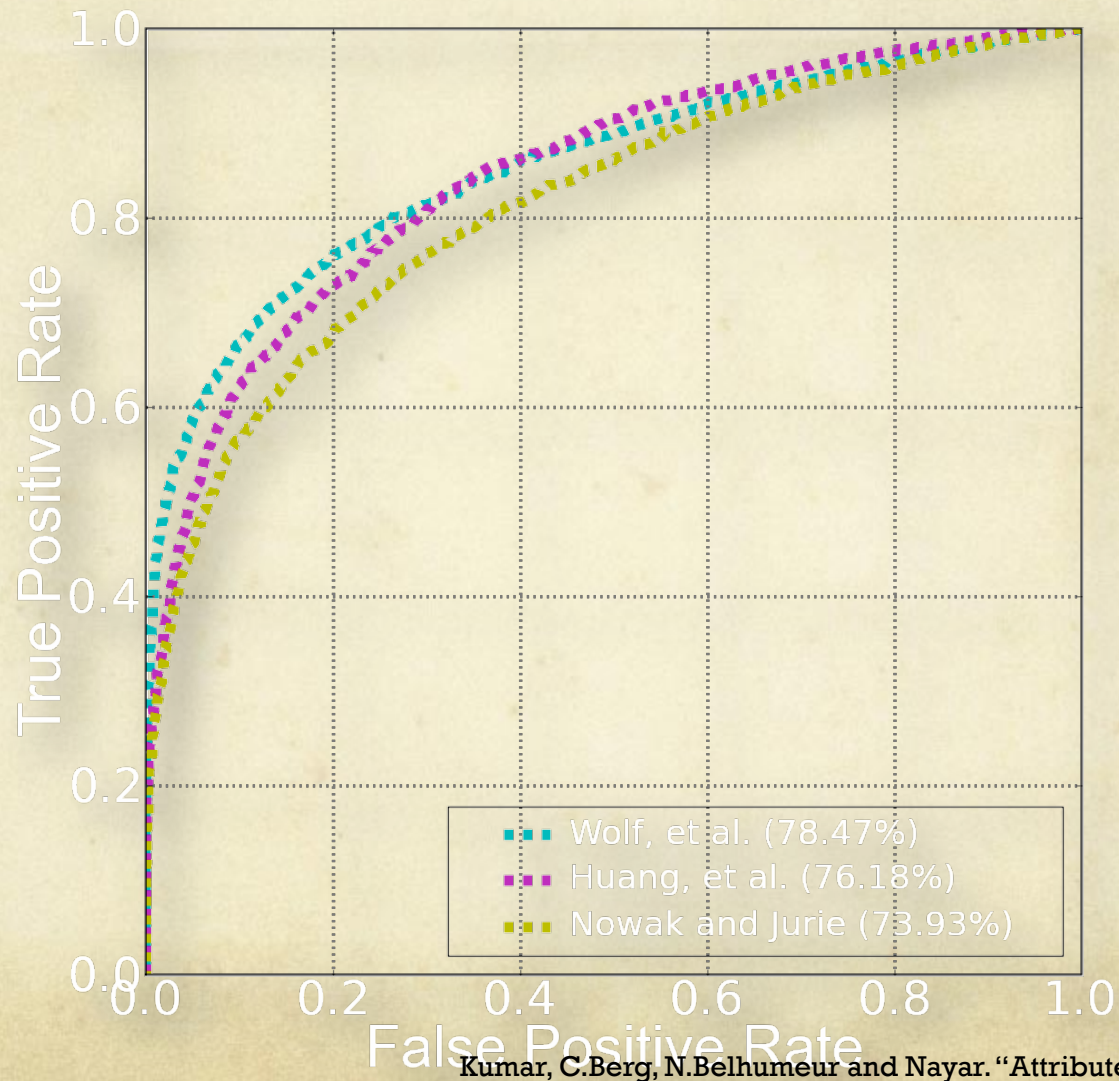
<http://vis-www.cs.umass.edu/lfw>

Kumar, C.Berg, N.Belhumeur and Nayar. "Attribute and
Simile Classifiers For Verification"

Experimental evaluation

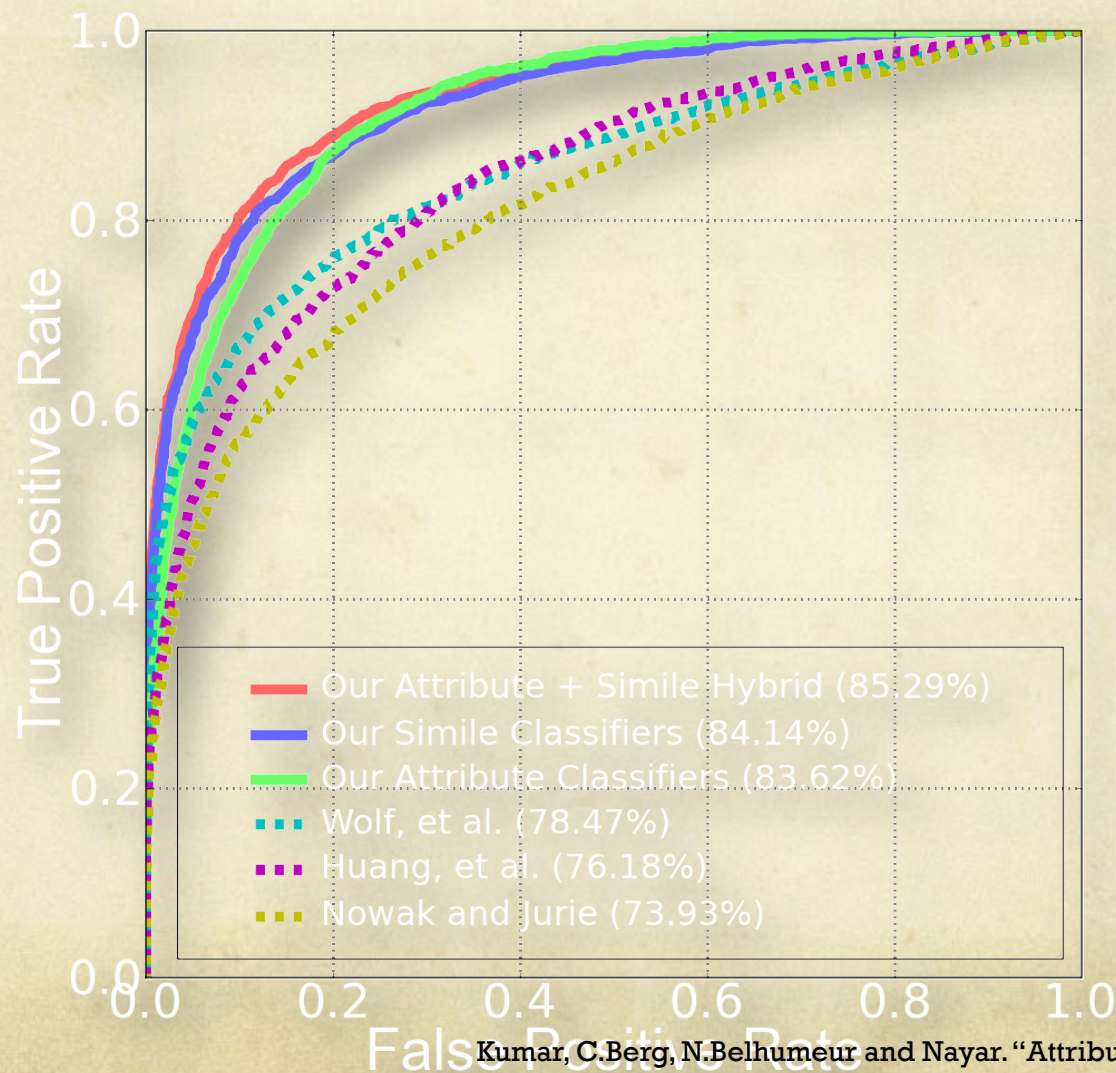
- LFW Image-Restricted Benchmark:
 - 6,000 face pairs (3,000 same, 3,000 different)
 - 10-fold cross-validation set of 6,000 pairs on which to evaluate the final performance

Previous Performance on LFW



Kumar, C.Berg, N.Belhumeur and Nayar. "Attribute and
Simple Classifiers For Verification"

Our Performance on LFW



Kumar, C. Berg, N. Belhumeur and Nayar. "Attribute and Simile Classifiers For Verification"

Human Face Verification Task

Are the two images of the same person or different?

Please click the button below each pair which best describes your confidence in whether the images are of the same person or not.

(It's fine to mark "maybe same" if you think they're the same person, but are not completely sure (similarly for different). If you cannot tell, please mark the middle "not sure" button.

- Once you make a choice, you can click "next pair" to go to the next pair.
- You can always go back to previous pairs if you want to change your answer.
- Once you finish all 30 pairs, there will be a submit button, which will complete the job.

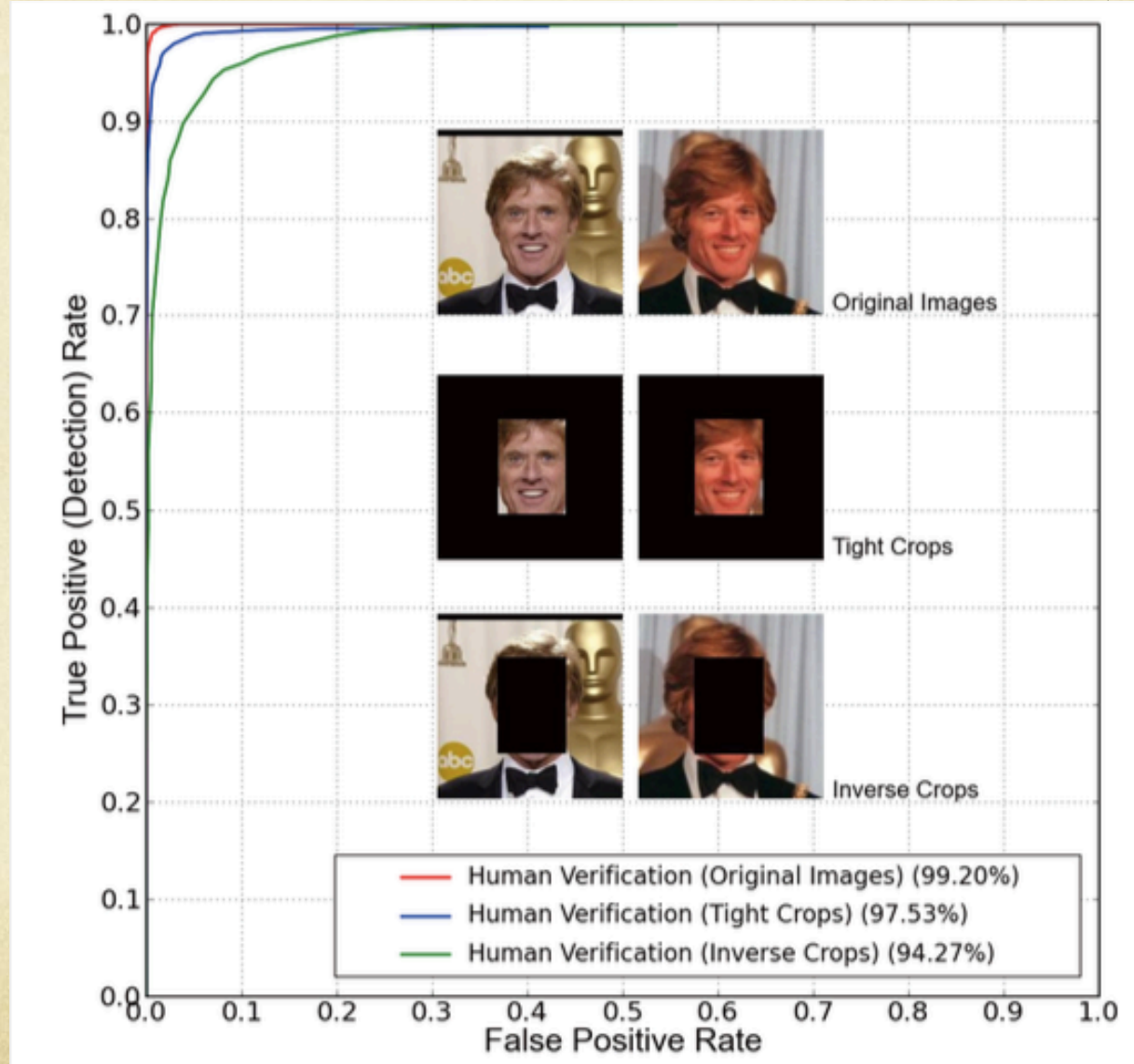
				
Definitely Same ☐	Maybe Same ☐	Not Sure ☒	Maybe Different ☐	Definitely Different ☐

Pair 1 of 30

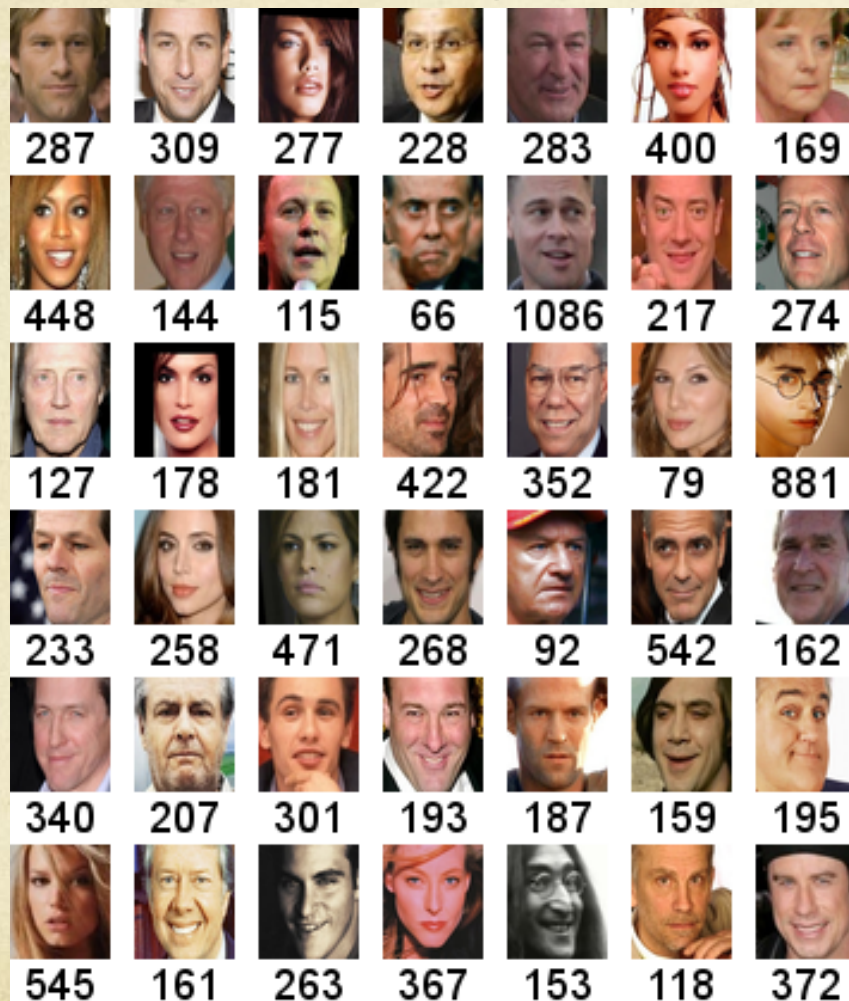
Kumar, C.Berg, N.Belhumeur and Nayar. "Attribute and
Simile Classifiers For Verification"

Next Pair

Human Performance on LFW



PubFig dataset & benchmark



Public figures:

- Politicians
- Celebrities

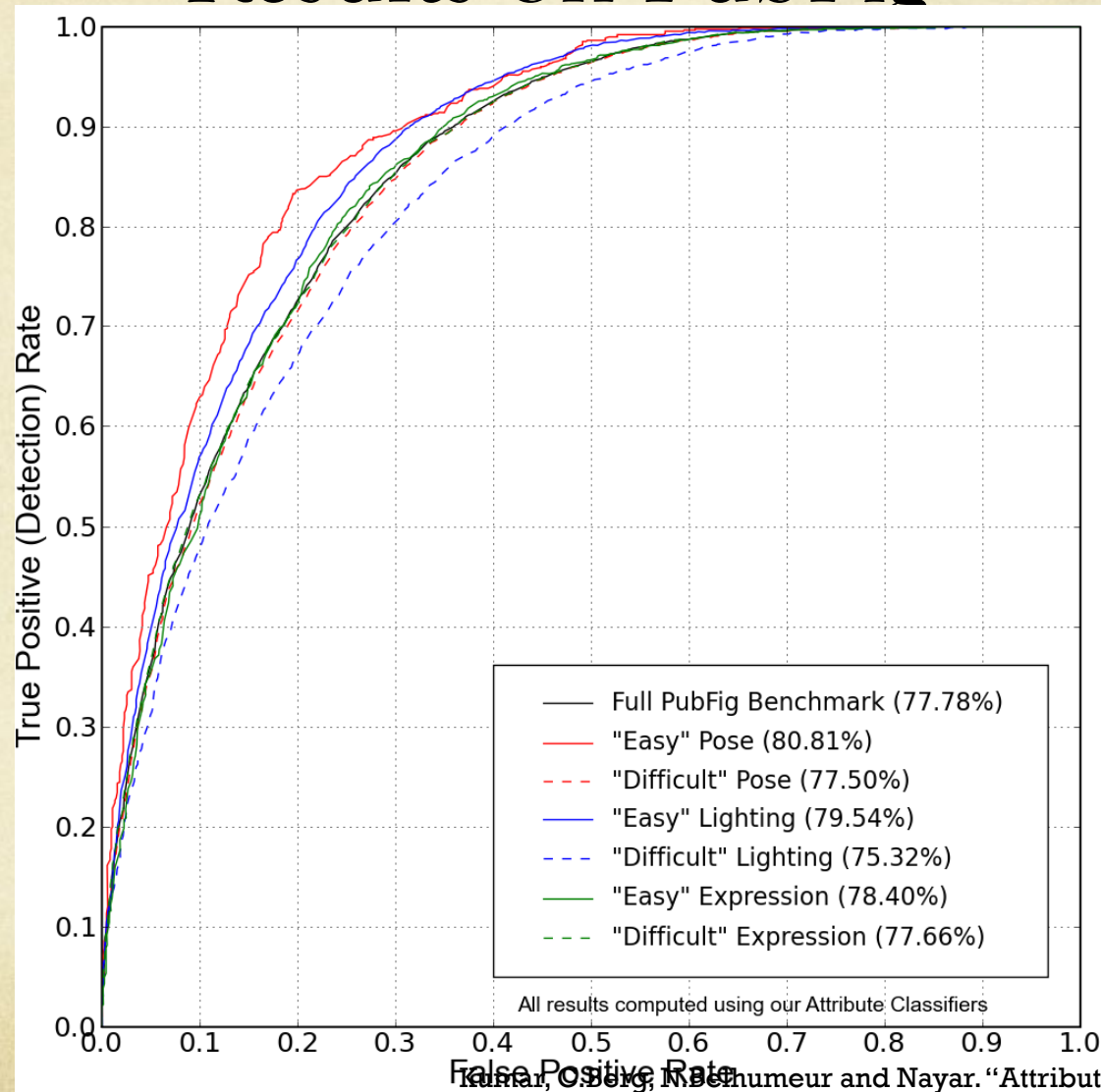
Larger & deeper:

- 60,000 Images
- 200 People
- 300 Images per person

Subsets:

- Pose
- Illumination
- Expression

Results on PubFig



Future Work

Q&A